Kestrel Carrier SLA Summary

Version 2026.3, effective 2026-09-01. Owner: @kestrel/platform-team. Source of truth for the commercial terms is each signed carrier agreement; this document summarises the operational terms engineers need. It is rendered to `carrier-sla.pdf` by `scripts/build_pdfs.py`.

1. Purpose and scope

Kestrel integrates with three carriers: NorthStar Lines (NSL), Baltic Express (BLX) and Pacific Reach (PCR). This document records the service levels each carrier commits to for the APIs Kestrel calls (capacity hold, confirmation, cancellation), the rate caps we must respect, and what Kestrel does in return. Engineers use it to set limits, size timeouts and decide when a carrier is "degraded" rather than "down".

2. Service levels at a glance

Term	NSL	BLX	PCR
Monthly API availability	99.5%	99.9%	99.0%
Capacity-hold p95 latency	800 ms	400 ms	1200 ms
Capacity-hold p99 latency	2000 ms	900 ms	3000 ms
Contractual request cap	40 rps	25 rps	15 rps
Burst allowance	60 rps for 10 s	35 rps for 5 s	none
Support hours	24x7	07:00 to 19:00 UTC	06:00 to 18:00 UTC
Planned maintenance	Sunday 01:00 to 03:00 UTC	Wednesday 22:00 to 23:00 UTC	first Saturday 00:00 to 04:00 UTC

The Kestrel booking service applies one limiter bucket per carrier (ADR-0007). The default rate, `kestrel.booking.ratelimit.per-carrier-rps` = 20, is below the NSL and BLX caps but above PCR's 15 rps, so PCR carries an override of 8 rps (`kestrel.booking.ratelimit.overrides.PCR`). Reviewing the other overrides is BOOK-418.

3. Availability and credits

Availability is measured by the carrier over calendar months using successful responses (any non-5xx response within the latency p99) divided by total requests, excluding planned maintenance and rate-limit rejections caused by Kestrel exceeding the cap.

Monthly availability (NSL)	Service credit
99.0% to below 99.5%	5% of monthly fees
98.0% to below 99.0%	10% of monthly fees
Below 98.0%	25% of monthly fees

BLX credits start below 99.9% (5%), below 99.5% (15%), below 99.0% (30%). PCR credits start below 99.0% (5%) and below 97.0% (20%). Kestrel must claim credits within 30 days of the month end; the platform team compiles evidence from booking metrics.

4. Rate limits and throttling

- Carriers return HTTP 429 when we exceed the cap. Sustained breach (more than 2 minutes above cap) lets the carrier suspend our credentials for up to 15 minutes.
- Kestrel must never retry a throttled call faster than the `Retry-After` the carrier sends.
- Kestrel's own limiter rejects early with `"carrier capacity exhausted"`; the gateway presents this to customers as `"booking temporarily unavailable"`.

- A new carrier's cap must be recorded here before go-live (see `runbooks/carrier-onboarding.md`).

5. Incident and support response

Carrier severity	Carrier acknowledges	Carrier target resolution
Critical (API down)	15 minutes (NSL), 30 minutes (BLX), 60 minutes (PCR)	4 hours
Major (degraded)	1 hour	8 hours
Minor	next business day	5 business days

Kestrel contact path: platform-team on-call opens the carrier incident through the carrier's operations portal, then emails the integration contact. Contact addresses are held in the shared password manager vault "Carrier contacts" (not in this repository). Integration contacts: NSL carrier-ops team, BLX integration desk, PCR partner support.

6. Security and credentials

- API credentials are issued per environment and rotated every 90 days with 24 hours of overlap (`runbooks/rotate-carrier-credentials.md`).
- Credentials are stored only in Vault under `secret/kestrel/carriers/<CODE>`.
- Carriers notify Kestrel within 72 hours of any security incident affecting our data.
- Kestrel notifies carriers within 24 hours of suspected credential exposure.

7. Data and booking semantics

- A capacity hold is valid for 30 minutes (NSL, BLX) or 20 minutes (PCR) unless confirmed.
- Duplicate holds for one shipment reference are billable as no-shows after the hold window: EUR 205 per hold for NSL and BLX, EUR 150 for PCR. Kestrel prevents these with the advisory lock in `BookingService.reserve()` (postmortem 2026-06-14).
- Cancellations within 2 hours of confirmation are free; later cancellations follow the carrier tariff.
- Amounts in statements are rounded half away from zero to the currency minor unit; this is the same rule as ADR-0005.

8. Maintenance and change notification

Carriers give at least 7 days' notice of planned maintenance and 30 days for API changes. Kestrel shifts traffic expectations accordingly: during NSL's Sunday window, booking errors for NSL are not a Kestrel incident unless they exceed the announced window.

9. Kestrel obligations

- Stay under the contractual request cap at all times.
- Pay invoices within 30 days; disputes raised within 45 days of the statement.
- Provide 60 days' notice of forecast volume increases above 25%.
- Keep integration contacts current in the vault.

10. Review

Reviewed each quarter by the platform-team with the booking-team. The next review is due 2026-12-01. History:

Version	Date	Change
2026.3	2026-09-01	Added limiter reference after ADR-0007
2026.2	2026-04-15	Updated PCR availability and credit terms
2026.1	2026-01-10	First consolidated summary